

Credit-Market Sentiment and the Business Cycle: A Replication of López-Salido et al. (2017)

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August 15, 2026

Abstract

We replicate the core results of López-Salido et al. (2017), who show that elevated credit-market sentiment forecasts a slowdown in real activity about two years ahead. We describe the data, present our own summary statistics, and reproduce every table and figure across the replication sample, an extended sample, and an Aaa-spread variant.

1 Overview

When credit is “frothy”—the López-Salido et al. (2017) thesis—spreads are narrow and low-quality issuance is high; those conditions mean-revert as activity turns down.

2 Data Sources

- **FRED** (Federal Reserve Bank of St. Louis): Baa/Aaa yields, Treasury yields, GDP. Several series are a newer vintage and must be stitched together; where the appendix is silent we made documented judgement calls.
- **Shiller** (Shiller, 2015): long-run market and P/E10 data.
- **Greenwood–Hanson high-yield share** (Greenwood and Hanson, 2013), pulled from the HBS workbook (Harvard Business School Behavioral Finance and Financial Stability Project) and spliced with Mergent FISD; a newer vintage than the paper.

3 Summary Statistics

Table 1: Descriptive statistics for the key series underlying the analysis. *Takeaway:* shows the scale, dispersion, and range of each variable so the reader can gauge skew and outliers before interpreting the regressions.

Variable	count	mean	std	min	p25	median	p75	max
Baa–Treasury spread (pp)	96	2.05	1.07	0.40	1.25	1.92	2.56	6.49
Aaa–Treasury spread (pp)	96	0.88	0.55	-0.11	0.39	0.80	1.25	2.63
Log high-yield share	95	-2.06	1.13	-6.02	-2.72	-1.89	-1.22	-0.45
Real GDP growth (%)	95	2.05	4.61	-14.46	0.62	2.09	3.93	16.21
S&P 500 return (%)	94	9.28	18.15	-51.58	-0.27	14.10	20.93	42.29
Log Shiller P/E10	95	2.85	0.43	2.06	2.46	2.85	3.15	3.79
Inflation (%)	96	3.04	3.79	-10.84	1.49	2.76	4.08	16.66
10y Treasury yield (%)	96	4.78	2.73	0.93	2.51	4.03	6.31	13.72
3m Treasury yield (%)	96	3.46	3.21	0.01	0.39	3.00	5.29	16.35
Change in credit spread (pp)	95	-0.01	0.83	-3.23	-0.39	-0.12	0.41	3.46

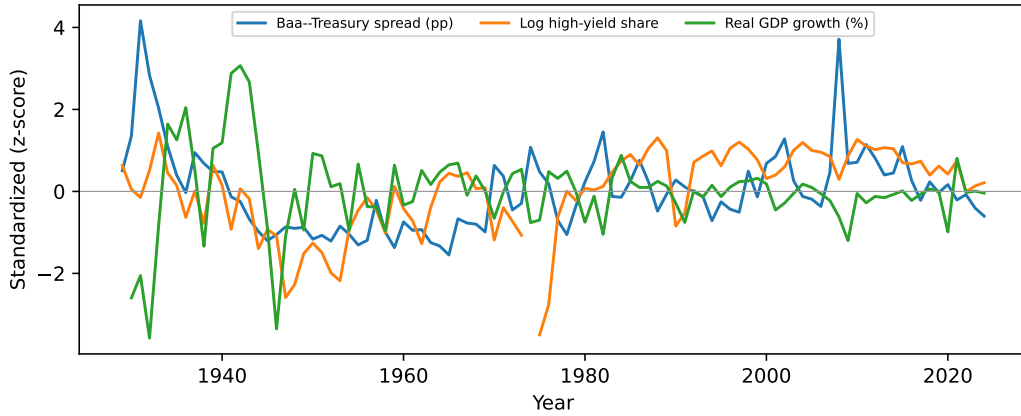


Figure 1: Standardized paths of the headline series. *Takeaway:* the credit spread, high-yield share, and GDP growth co-move over the cycle, previewing the mechanism the regressions formalize.

4 Replication Sample (1929–2015)

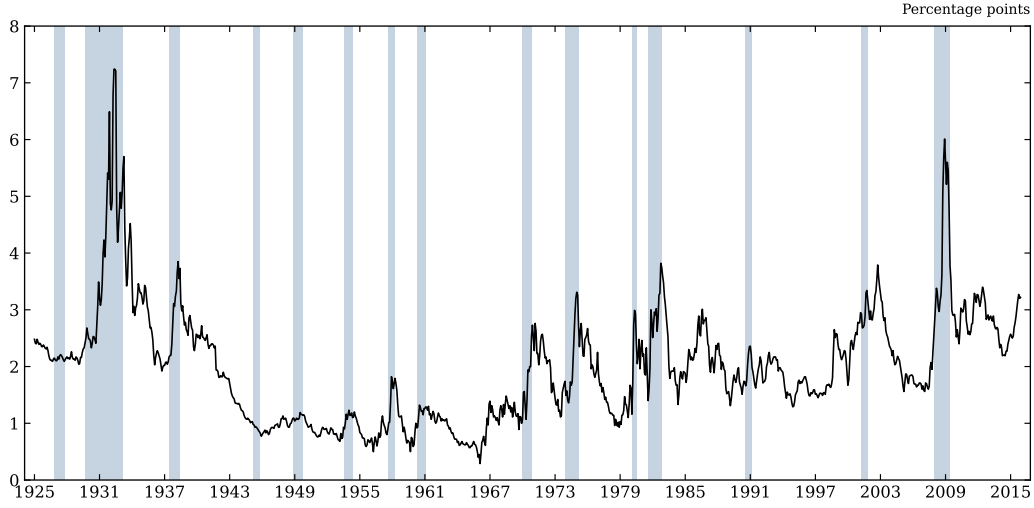


Figure 2: Figure I. Baa–Treasury credit spread with NBER recessions shaded. *Takeaway:* the spread spikes into every recession, motivating it as a cyclical signal.

Table 2: Table I. One-year-ahead GDP growth on the lagged credit-spread change vs. equity returns. *Takeaway:* the credit term is significant where equity returns are not.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.958*** (0.585)	—	-2.124*** (0.578)
r_t^{SP}	—	0.075** (0.033)	0.022 (0.035)
Δy_{t-1}	0.476*** (0.104)	0.478*** (0.136)	0.464*** (0.114)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.248 (0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.814** (0.360)
π_{t-1}	—	—	0.095 (0.076)
$\bar{R}^2$	0.426	0.381	0.453
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.365	—	-0.396
r_t^{SP}	—	0.299	0.087

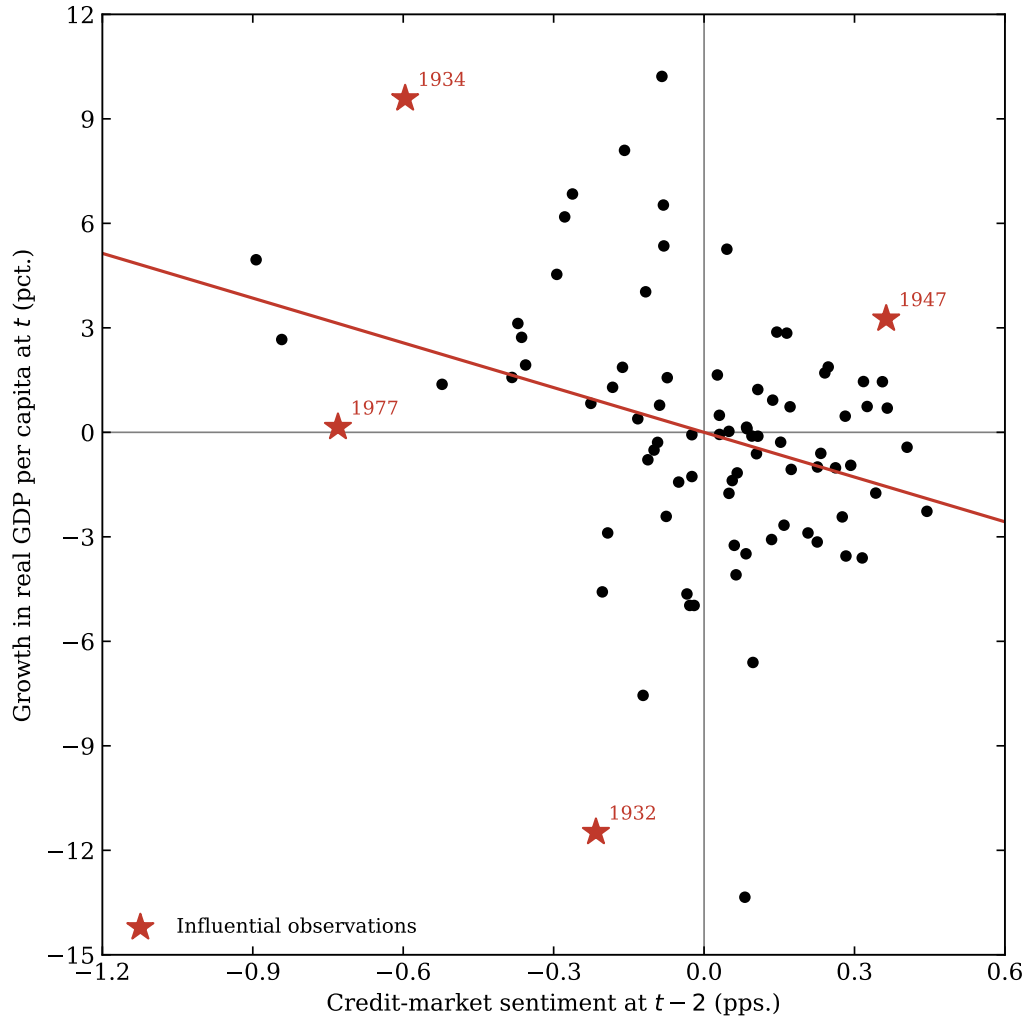


Figure 3: Figure II. Credit-market sentiment vs. subsequent growth. *Takeaway:* the downward slope shows frothy credit precedes weaker growth.

Table 3: Table II. Sentiment-based predictive regressions with the auxiliary decomposition. *Takeaway:* lagged log high-yield share and spread jointly forecast spread changes.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.281*** (1.253)	—	-3.861*** (1.301)	-4.907*** (1.678)
$\hat{r}_t^{SP}$	—	0.145* (0.082)	0.053 (0.074)	—
Δy_{t-1}	0.597*** (0.114)	0.534*** (0.099)	0.590*** (0.112)	0.581*** (0.096)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.099 (0.313)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.616 (0.414)
π_{t-1}	—	—	—	0.123 (0.154)
R^2	0.377	0.337	0.380	0.394
Auxiliary regressions		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.147*** (0.042)	—	
s_{t-2}		-0.245*** (0.047)	—	
$\ln[P/E10]_{t-2}$		—	-0.132*** (0.040)	
R^2		0.102	0.083	

5 Extended Sample (1929–2025)

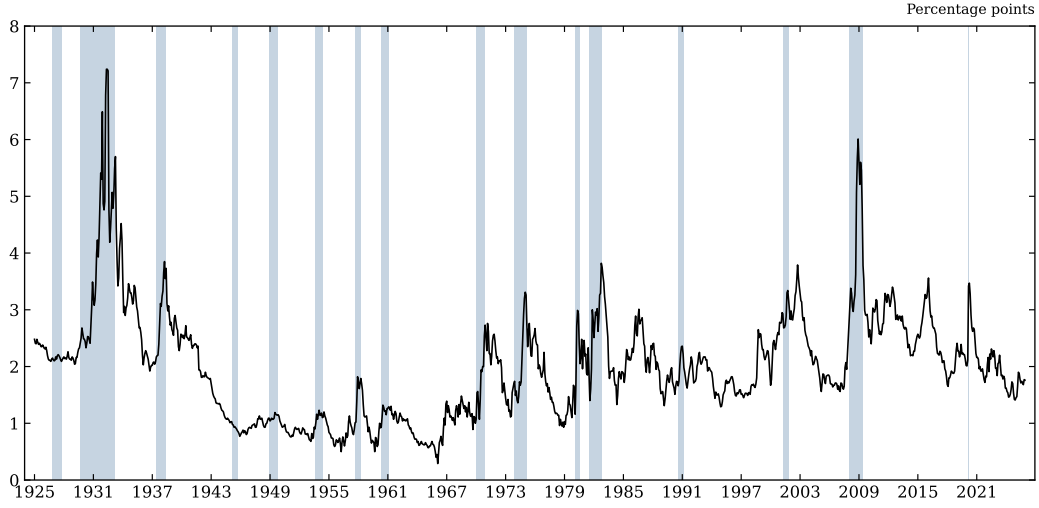


Figure 4: Figure I, extended. *Takeaway:* the cyclical pattern persists through recent data.

Table 4: Table I, extended sample. *Takeaway:* the credit result survives out of the original window.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.874*** (0.566)	—	-2.020*** (0.572)
r_t^{SP}	—	0.070** (0.033)	0.018 (0.035)
Δy_{t-1}	0.461*** (0.108)	0.464*** (0.136)	0.450*** (0.116)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.183 (0.269)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.752** (0.367)
π_{t-1}	—	—	0.088 (0.078)
$\bar{R}^2$	0.395	0.354	0.415
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.350	—	-0.378
r_t^{SP}	—	0.286	0.072

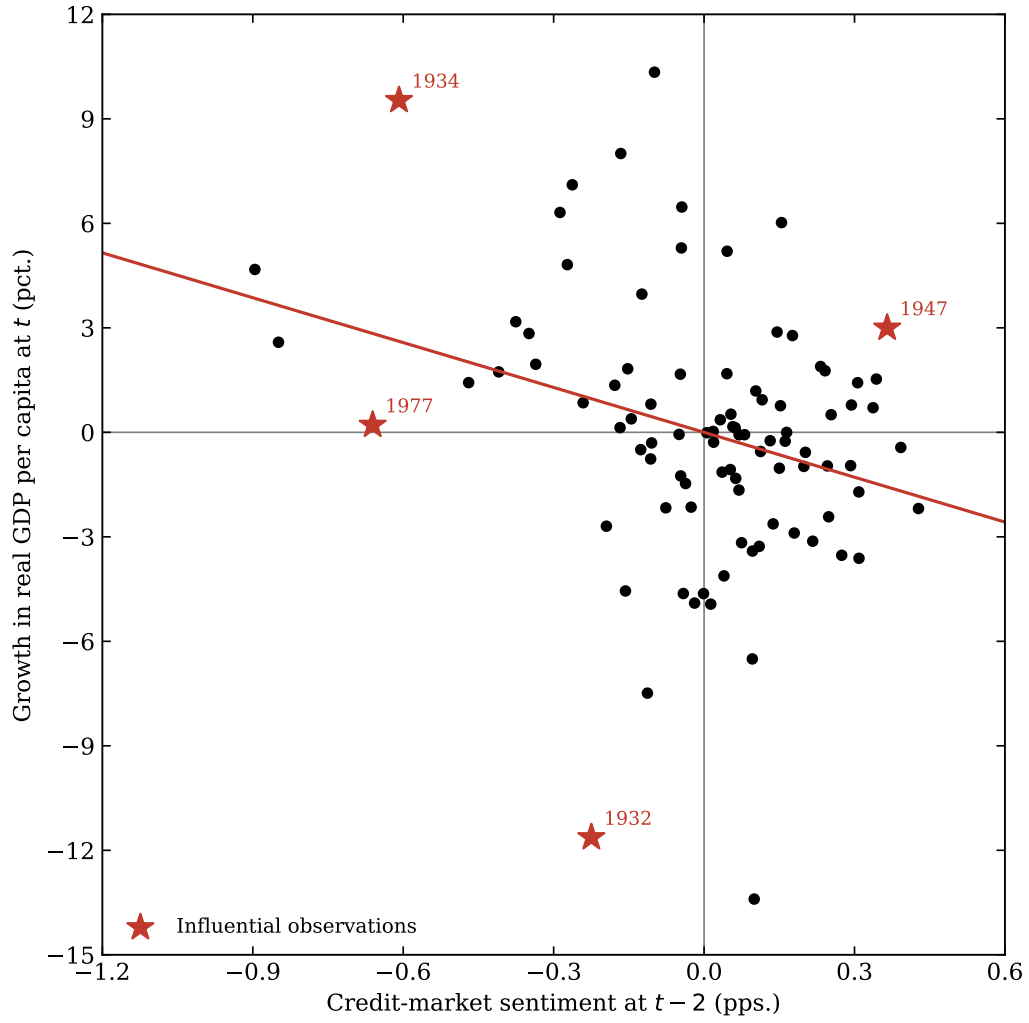


Figure 5: Figure II, extended. *Takeaway:* the negative sentiment–growth relation holds up.

Table 5: Table II, extended sample. *Takeaway:* coefficients are close to the replication values.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.294*** (1.295)	—	-3.901*** (1.316)	-4.943*** (1.703)
$\hat{r}_t^{SP}$	—	0.158* (0.092)	0.062 (0.081)	—
Δy_{t-1}	0.577*** (0.118)	0.516*** (0.103)	0.570*** (0.115)	0.559*** (0.100)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.123 (0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.612 (0.410)
π_{t-1}	—	—	—	0.121 (0.154)
R^2	0.353	0.313	0.356	0.368
Auxiliary regressions				r_t^{SP}
$\ln \text{HYS}_{t-2}$		0.129*** (0.039)	—	
s_{t-2}		-0.243*** (0.046)	—	
$\ln[P/E10]_{t-2}$		—	-0.103*** (0.034)	
R^2		0.095	0.059	

6 Extension: Aaa Credit Spread

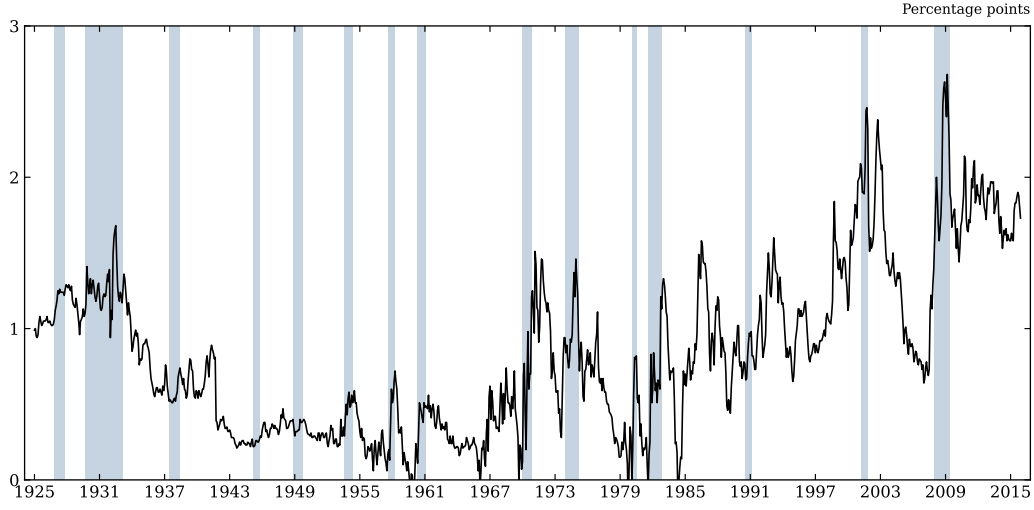


Figure 6: Figure I, Aaa (replication). *Takeaway:* the higher-grade spread is smoother but still cyclical.

Table 6: Table I, Aaa (replication). *Takeaway:* the Aaa spread carries weaker but same-signed predictive content.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.958*** (0.734)	—	-3.192*** (1.168)
r_t^{SP}	—	0.075** (0.033)	0.056 (0.035)
Δy_{t-1}	0.517*** (0.109)	0.478*** (0.136)	0.477*** (0.121)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.326 (0.314)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.917* (0.504)
π_{t-1}	—	—	0.102 (0.108)
$\bar{R}^2$	0.321	0.381	0.396
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.165	—	-0.269
r_t^{SP}	—	0.299	0.225

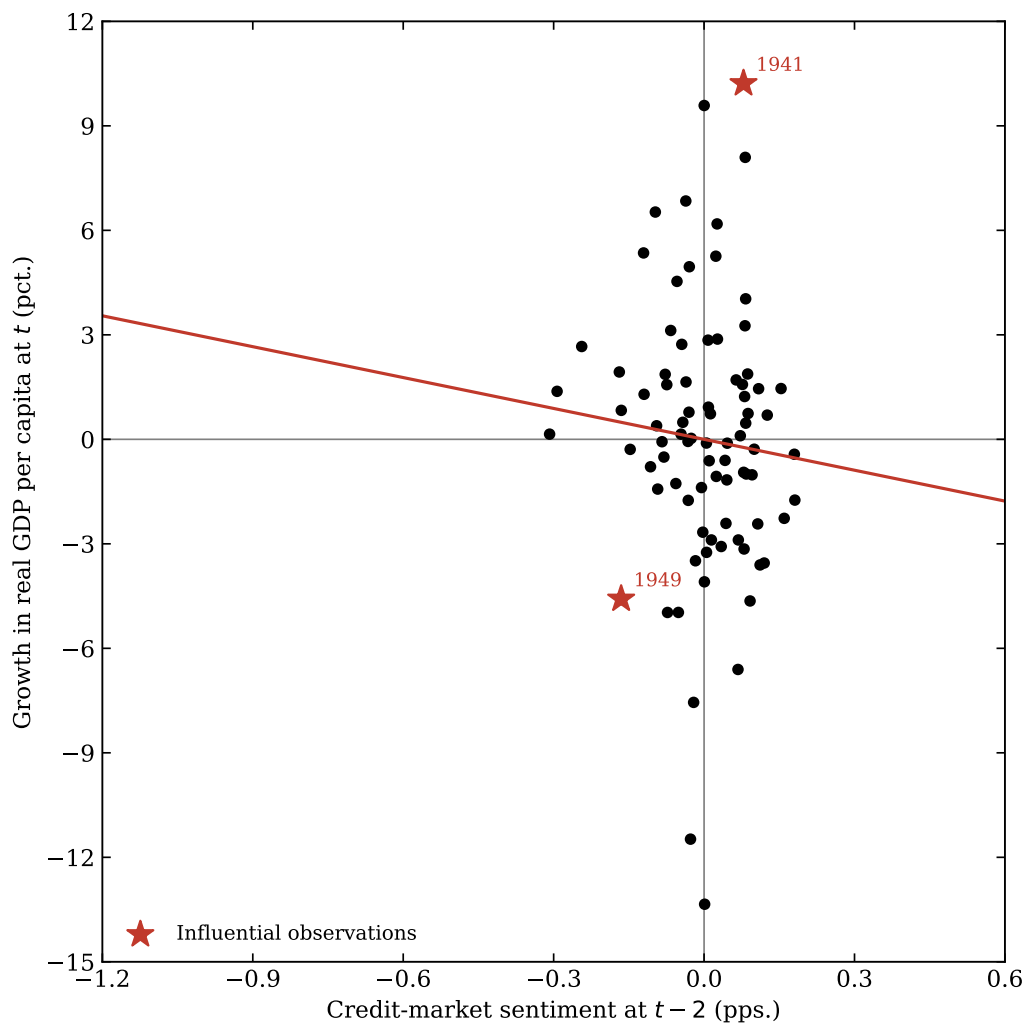


Figure 7: Figure II, Aaa (replication). *Takeaway:* sentiment built on Aaa shows a flatter slope.

Table 7: Table II, Aaa (replication). *Takeaway:* the auxiliary decomposition is weaker for the higher-grade spread.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.956 (2.488)	—	-2.362 (2.865)	-3.663 (2.806)
$\hat{r}_t^{SP}$	—	0.145* (0.082)	0.131 (0.085)	—
Δy_{t-1}	0.555*** (0.108)	0.534*** (0.099)	0.549*** (0.100)	0.559*** (0.105)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.063 (0.308)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.373 (0.394)
π_{t-1}	—	—	—	0.043 (0.132)
R^2	0.321	0.337	0.342	0.328

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \text{HYS}_{t-2}$	0.078*** (0.029)	—
s_{t-2}	-0.167*** (0.058)	—
$\ln[P/E10]_{t-2}$	—	-0.132*** (0.040)
R^2	0.064	0.083

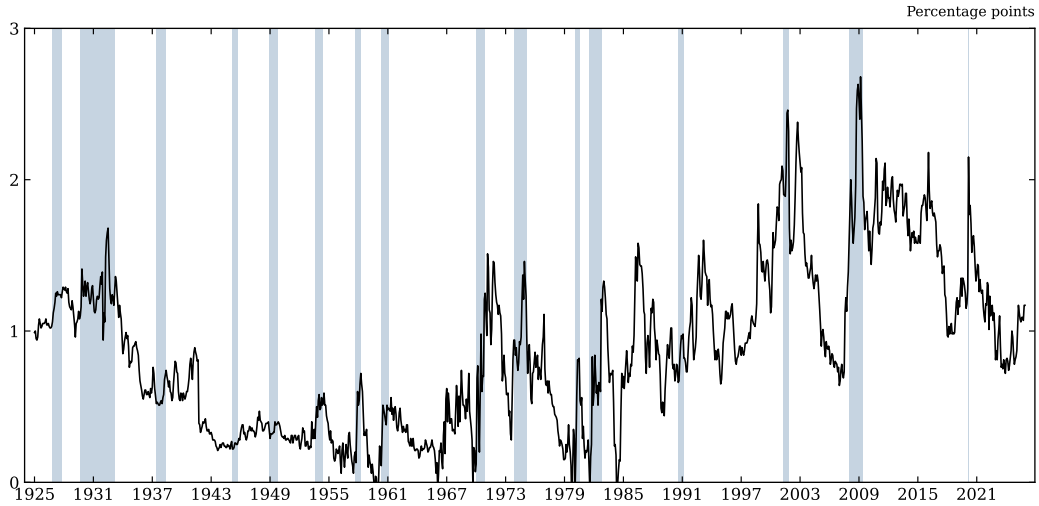


Figure 8: Figure I, Aaa (extended). *Takeaway:* the Aaa pattern extends to the present.

Table 8: Table I, Aaa (extended). *Takeaway:* extended Aaa estimates remain same-signed.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.810** (0.714)	—	-2.873** (1.165)
r_t^{SP}	—	0.070** (0.033)	0.052 (0.035)
Δy_{t-1}	0.500*** (0.112)	0.464*** (0.136)	0.463*** (0.124)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.257 (0.289)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.797 (0.495)
π_{t-1}	—	—	0.091 (0.109)
$\bar{R}^2$	0.297	0.354	0.360
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.154	—	-0.244
r_t^{SP}	—	0.286	0.210

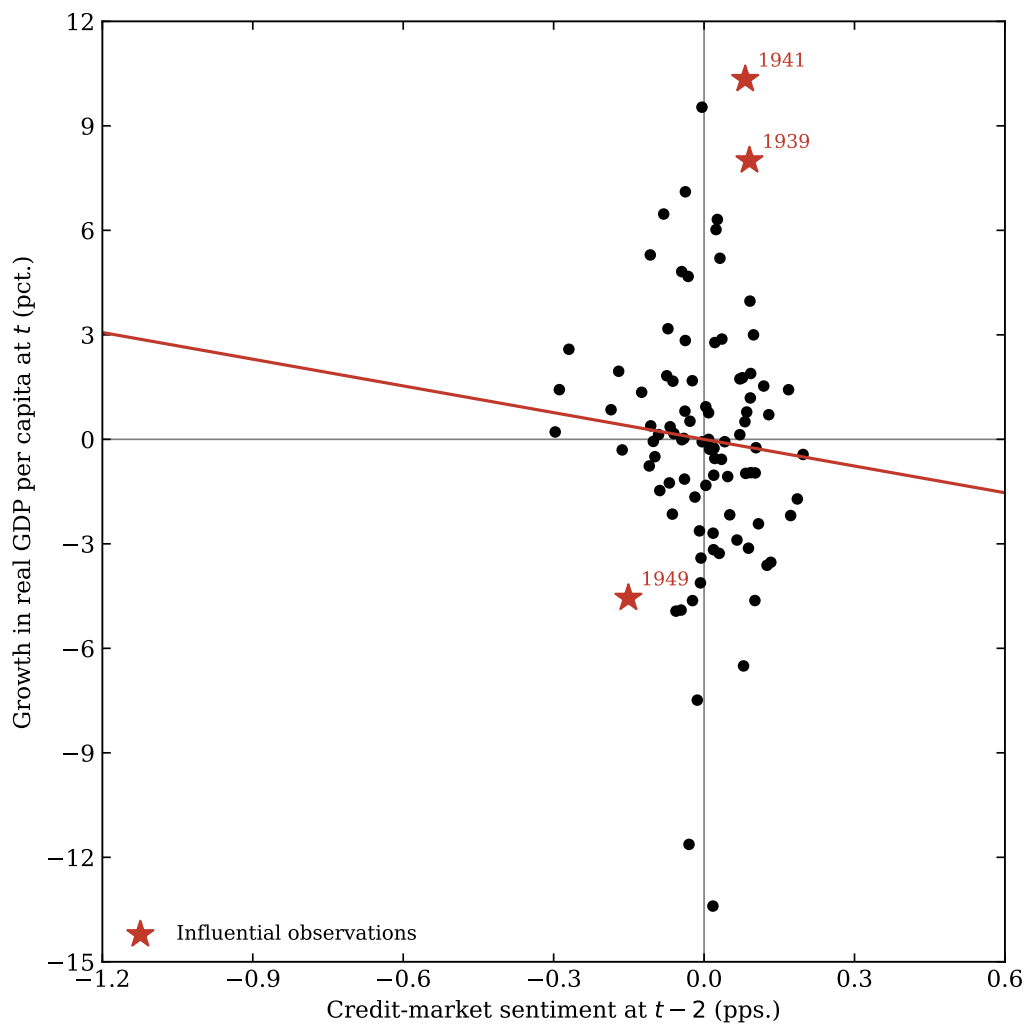


Figure 9: Figure II, Aaa (extended). *Takeaway:* the relation weakens but does not reverse.

Table 9: Table II, Aaa (extended). *Takeaway:* consistent with the replication-sample Aaa result.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.556 (2.389)	—	-2.546 (2.578)	-3.219 (2.717)
$\hat{r}_t^{SP}$	—	0.158* (0.092)	0.146 (0.094)	—
Δy_{t-1}	0.536*** (0.111)	0.516*** (0.103)	0.532*** (0.103)	0.540*** (0.107)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.050 (0.286)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.367 (0.389)
π_{t-1}	—	—	—	0.041 (0.132)
R^2	0.298	0.313	0.318	0.306
Auxiliary regressions		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.076*** (0.028)	—	
s_{t-2}		-0.182*** (0.058)	—	
$\ln[P/E10]_{t-2}$		—	-0.103*** (0.034)	
R^2		0.069	0.059	

7 Successes and Challenges

The long-run spread reproduces cleanly and the credit-beats-equity result holds ($R^2 \approx 0.38$). The main difficulties were sourcing the high-yield share and matching the paper’s Newey and West (1987) standard errors; vintage and series-stitching choices explain small differences from the published numbers.

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